

ANOVA (Analysis of Variance)

Introduction

ANOVA (Analysis of Variance) is a statistical hypothesis test used to compare the *means of three or more independent groups* to determine whether there is a statistically significant difference among them.

Instead of performing multiple t-tests (which increase the probability of Type I error), ANOVA uses a *single F-test* to compare all groups simultaneously.

Analysis of Variance (ANOVA) is a statistical technique that compares the variation **between groups** with the variation **within groups**.

If the variation between groups is much larger than the variation within groups, the group means are considered significantly different.

Objectives of ANOVA

- Compare the means of three or more populations.
- Determine whether observed differences are statistically significant.
- Minimize Type I error that occurs with multiple t-tests.
- Identify whether different treatments produce different results.

Types of ANOVA

Type	Purpose
One-Way ANOVA	One independent variable with three or more groups
Two-Way ANOVA	Two independent variables
Repeated Measures ANOVA	Same subjects measured multiple times
MANOVA	Multiple dependent variables

Assumptions of ANOVA

Before applying ANOVA, the following assumptions must be satisfied:

1. Independence

Each observation must be independent.

2. Normality

Each population should follow a normal distribution.

3. Equal Variance

Population variances should be approximately equal.

Hypotheses

Suppose there are **k groups**.

Null Hypothesis

$$H_0: \mu_1 = \mu_2 = \mu_3 = \dots = \mu_k$$

All population means are equal.

Alternative Hypothesis

$$H_1: \text{At least one mean differs}$$

Terminologies must know before ANOVA test

Population Mean

$$\mu$$

Average value of an entire population.

Sample Mean

$$\bar{X}$$

Average of sample observations.

Grand Mean

Average of all observations from every group.

$$\bar{X}_G = \frac{\text{Total of all observations}}{\text{Total number of observations}}$$

Sources of Variation

ANOVA divides total variation into two parts.

A. Between-Group Variation

Variation caused by differences between group means.

Example:

Class A average = 80

Class B average = 90

Class C average = 70

These differences contribute to **between-group variation**.

B. Within-Group Variation

Variation among observations inside the same group.

Example

Class A

78,80,81,79

Even within Class A, marks differ.

This contributes to **within-group variation**.

Formulae Used in ANOVA

Grand Mean

$$\bar{X}_G = \frac{\sum X}{N}$$

Sum of Squares Between Groups (SSB)

$$SSB = \sum n_i (\bar{X}_i - \bar{X}_G)^2$$

Where

- (n_i) =Number of observations in each group
- $(\bar{X}_i)$ =Group mean
- $(\bar{X}_G)$ =Grand mean

Sum of Squares Within Groups (SSW)

$$SSW = \sum (X - \bar{X}_i)^2$$

Total Sum of Squares

$$SST = SSB + SSW$$

Degrees of Freedom

Between Groups

$$df_B = k - 1$$

Within Groups

$$df_W = N - k$$

Total

$$df_T = N - 1$$

Mean Square

Between Groups

$$MSB = \frac{SSB}{df_B}$$

Within Groups

$$MSW = \frac{SSW}{df_w}$$

F Statistic

$$F = \frac{MSB}{MSW}$$

ANOVA Procedure

Step 1

State hypotheses.

Step 2

Choose significance level

Usually

$$\alpha = 0.05$$

Step 3

Calculate each group mean.

Step 4

Calculate Grand Mean.

Step 5

Calculate SSB.

Step 6

Calculate SSW.

Step 7

Calculate SST.

Step 8

Find degrees of freedom.

Step 9

Calculate Mean Squares.

Step 10

Calculate F-value.

Step 11

Compare with F-table.

Step 12

Draw conclusion.

Example Question

Three teachers use different teaching methods.

The marks obtained by students are:

Teacher A	Teacher B	Teacher C
12	18	24
15	20	22
14	19	23

Test whether the average marks are significantly different at the **5% level**.

Step 1

Hypotheses

$$H_0: \mu_1 = \mu_2 = \mu_3$$

All teaching methods have the same average marks.

$$H_1: \text{At least one mean differs}$$

Step 2

Level of significance

$$\alpha = 0.05$$

Step 3

Calculate Means

Teacher A

$$\bar{X}_A = \frac{12 + 15 + 14}{3} = \frac{41}{3} = 13.67$$

Teacher B

$$\bar{X}_B = \frac{18 + 20 + 19}{3} = \frac{57}{3} = 19$$

Teacher C

$$\bar{X}_C = \frac{24 + 22 + 23}{3} = \frac{69}{3} = 23$$

Step 4

Grand Mean

Total observations

$$N = 9$$

Total marks

$$41 + 57 + 69 = 167$$

Grand Mean

$$\bar{X}_G = \frac{167}{9} = 18.56$$

Step 5

Calculate SSB

Formula

$$SSB = \sum n (\bar{X}_i - \bar{X}_G)^2$$

Since each group has

$$n = 3$$

Teacher A

$$3(13.67 - 18.56)^2 = 3(4.89)^2 = 71.72$$

Teacher B

$$3(19 - 18.56)^2 = 0.58$$

Teacher C

$$3(23 - 18.56)^2 = 59.16$$

Therefore

$$SSB = 71.72 + 0.58 + 59.16 = 131.46$$

Step 6

Calculate SSW

$$SSW = \sum (X - \bar{X}_i)^2$$

Teacher A

$$(12 - 13.67)^2 + (15 - 13.67)^2 + (14 - 13.67)^2$$

$$= 2.79 + 1.77 + 0.11 = 4.67$$

Teacher B

$$(18 - 19)^2 + (20 - 19)^2 + (19 - 19)^2$$

$$= 1 + 1 + 0 = 2$$

Teacher C

$$(24 - 23)^2 + (22 - 23)^2 + (23 - 23)^2$$

$$= 1 + 1 + 0 = 2$$

Therefore

$$SSW = 4.67 + 2 + 2 = 8.67$$

Step 7

Total Sum of Squares

$$SST = SSB + SSW$$

$$SST = 131.46 + 8.67 = 140.13$$

Step 8

Degrees of Freedom

Number of groups

$$k = 3$$

Total observations

$$N = 9$$

Between

$$df_B = 3 - 1 = 2$$

Within

$$df_W = 9 - 3 = 6$$

Total

$$df_T = 9 - 1 = 8$$

Step 9

Mean Squares

Between

$$MSB = \frac{131.46}{2} = 65.73$$

Within

$$MSW = \frac{8.67}{6} = 1.45$$

Step 10

Calculate F

$$F = \frac{65.73}{1.45} = 45.33$$

Step 11

Critical Value

From the F-table

$$df_1 = 2$$

$$df_2 = 6$$

At

$$\alpha = 0.05$$

$$F_{critical} = 5.14$$

Step 12

Decision

Calculated value: 45.33

Critical value

5.14

Since

$$45.33 > 5.14$$

Reject the null hypothesis.

Step 13

Conclusion

There is **sufficient evidence** to conclude that the average marks obtained under the three teaching methods are **not all equal**. At least one teacher's teaching method results in a significantly different mean score.

Practice Problems

Example 1

Compare the average yields (kg) of crops using three fertilizers:

- Fertilizer A: 25, 28, 27, 26
- Fertilizer B: 30, 29, 31, 32
- Fertilizer C: 24, 23, 25, 22

Use one-way ANOVA at the 5% significance level.

Example 2

Three machines produce the following output (units/hour):

- Machine A: 100, 102, 101, 99
- Machine B: 95, 97, 96, 94
- Machine C: 103, 105, 104, 102

Test whether the mean production differs significantly among the three machines.

Example 3

A researcher compares the average test scores of students taught using three teaching methods:

- Method A: 65, 68, 70, 66
- Method B: 72, 75, 74, 73
- Method C: 69, 71, 70, 68

Perform a one-way ANOVA to determine whether the teaching methods produce different average scores.